The Inductive Biases of Stochastic Gradient Descent

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1 Introduction

If SGD approximates Bayesian inference, we can leverage the rich literature on Bayesian statistics to obtain more rigorous asymptotic guarantees for AI safety. For instance the local learning coefficient from singular learning theory enables us to understand the asymptotic behavior of in-distribution generalization error in deep neural networks by measuring the model's effective complexity. Furthermore, we can derive higher-order statistical moments of the distribution using the model's free energy. Additionally, being able to sample from the distribution of SGD trajectories would prove valuable for assessing the robustness of some safety property. More specifically, we could sample models from the underlying distribution of parameters within some degenerate subspace of parameters that are compatible with minimizing the loss and evaluate their safety properties. If a significant proportion of sampled models exhibit unsafe behaviors (such as deception), this would provide evidence that the corresponding training checkpoint is not robuslty safe.

The assumption that SGD approximates Bayesian inference holds when we can model SGD as Langevin dynamics and take the long time limit of SGD (Mandt 2018). This regime requires several conditions: the data must be i.i.d., the batch size must be large enough to invoke the central limit theorem, the learning rate must be small, and the noise covariance must be isotropic, finite, and non degenerate. Under these conditions, we can solve the Fokker-Planck equation of the Langevin dynamics to show that the long-run distribution of SGD converges to the tempered posterior (Gibbs distribution). The corresponding Bayesian posterior is the special case where temperature $\mathrm{T} = 1$. However, this regime does not accurately describe actual deep neural network training. Specifically, standard assumptions for the Langevin dynamics that are not satisfied:

- **Gradient Noise Anisotropy**: Gradient noise covariance depends on the geometry of the loss surface. In particular, it is sensitive to degeneracies in the loss surface which add a stickiness effect to SGD dynamics. Furthermore, the gradient noise covariance is not always invertible due to degenerate components in the loss landscape.
- **Loss landscape is degenerate** and the gradient noise covariance is non invertible at a degenerate critical point.
- **Heavy-tailed Gradient Noise**: Empirically violated due to observed heavy-tailed gradient noise distributions. The Wiener process in the SDE model of SGD could be replaced by some α-stable process.
- **Detailed Balance (stationary distribution at equilibrium with Curl-Free Probability Current)**: In general we expect a non-zero curl in the probability currents for the stationary distribution.

- **Small Learning Rate Approximation**: Empirically having a finite non infinitesimal learning rate is important as can be seen by edge-of-stability phenomena with oscillatory dynamics.

These violations suggest that the distribution of SGD trajectories deviates from the tempered posterior resulting from Langevin models of SGD.

To address these shortcomings, we propose investigating refined SDE models of SGD that incorporate key empirical phenomena:

- **Anisotropic Covariance**: Models the anisotropic gradient noise empirically observed during training.
- **Heavy-tailed Noise generated by α-stable Distributions**: Captures the heavy-tailed gradient noise distributions observed empirically during training.
- **Central Flow Term**: Accounts for oscillatory dynamics and sharpness-induced flows arising from edge-of-stability phenomena (see the Central Flow paper).
- **Fractional Time Fokker-Planck Derivative**: Describes the subdiffusive behavior of SGD (see Max Hennick's paper on fractal dynamics).
- **Variant of SGD with momentum and RMSprop (e.g. ADAM)**: Adam optimizers are used in LLMs training, not SGD. Langevin must be modified into an underdamped dynamics with momentum (only take momentum into account).

Some key questions

Another perspective here is what are the key guiding questions of our investigation? Mirroring the implicit biases of gradient descent, we want to understand:

- Under what conditions using Bayesian inference as a model of training dynamics is valid?
- Under what conditions can we use the continuous time limit of GD as a model of training dynamics?
- When is taking into account the noise anisotropy important?
- When do we need to take into account the heavy-tailed noise?
- When is the SDE persepctive on training dynamics useful, in particular when does it makes sense to look at the stationary or quasi-stationary distribution?
- How much does it matter that we consider GD, SGD, ADAM, RMSprop etc?

Connection to AI safety

Another important direction would be to connect the study of the inductive biases of SGD with AI alignment. In particular, we want to get a clearer idea of what theoretical results would be particularly useful for scalable oversight (e.g. debate) or other safety agendas (e.g. ARC, interpretability). At the moment there are some broad directions that seem promising but I want to get a clearer picture of the safety relevance and time sensitivity of applications to AI alignment.

- **Robustness of some safety property**: With an accurate sampler of the distribution of parameters selected by SGD, we could sample models compatible with minimizing the loss and evaluate their safety propertie to test their robustness (e.g. by sampling quasi-stationary distribution).

- **Defining new observables to track during training that are analogous to SLT** but adapted to SGD and its variants. For example, we could compute a Free energy of the (out of equilibrium) distribution of parameters selected by SGD during training and derive a measure of effective complexity and other statistical moments. A particularly interesting result here would be computing the in-distribution, and out-of-distribution generalization error in terms of these statistical quantities.
- **Understanding better how SGD matters for learning algorithms during training relative to Bayes**, in particular relating the inductive biases of SGD to learning algorithms step by step relative to Bayes posterior which directly "teleports" to a solution.
- **Broad safety applications**: How we can use this to include or exclude some data to make the training process more likely to converge to a safe solution.

In terms of the theory of change there are two important failure modes to avoid in this project:

- Doing work that is better to delegate to future AI as I expect they'll accelerate theory work a lot. So we want to focus on theory work that is time-sensitive.
- Doing work that is fundamental but difficult to relate to alignment for the next couple of years and that is better suited to academics already working on theory of deep learning.

2 Setup and mystery of deep learning

What to introduce: - The problem of learning a function from data - Dataset, input, output, i.i.d assumption - Population loss, empirical loss, generalization error - Want to do argmin of the loss function. - Learning as finding hypothesis that minimize the population loss and empirical loss - Bound over generalization error - Neural networks as a particular function class - Naive complexity bounds suggest that learning should be very difficult (curse of dimensionality) - In general not possible because argmin is intractable? - Instead we do some local search, A typical choice is using GD. - Neural networks are successful in practice. There is some implicit regularization at play during training. Understanding training dynamics to make this implicit regularization explicit is key. - We know from SLT that degeneracies play an important role: implicit model selection during Bayesian inference with LLC as complexity measure. Give tighter bounds on generalization.

Let $D_m = ((X_i, Y_i))_{i=1}^m$ be a dataset of m i.i.d. samples from some unknown distribution P. Let $f : \mathcal{X} \to \mathcal{Y}$ be a function. A generic problem in (supervised) machine learning is to learn a function h in a set of hypotheses class $\mathcal{H}$ that best approximates f. More specifically, we want to find a function h in $\mathcal{H}$ that minimizes the population loss [1]:

$$L(h) := \mathbb{E}_{(X,Y)\sim P}[||Y - h(X)||_2] \tag{1}$$

The population loss is the expected loss over the unknown distribution P. In practice, the learner does not have access to the unknown distribution P but only to a finite dataset D_m. Instead of minimizing the population loss, the learner minimizes the empirical loss which is the average error over the dataset D_m:

$$L_m(h; D_m) = \frac{1}{m}\sum_{i=1}^{m} ||y_i - h(x_i)||_2 \tag{2}$$

[1] For simplicity of the exposition we consider the mean squared error loss

The law of large numbers ensures that the empirical loss converges to the population loss as the number of samples m goes to infinity. In other words, in the limit of large samples the empirical can be used to approximate the population loss. The problem of approximating the function f is framed as an optimization problem over a set of hypotheses $\mathcal{H}$ given some data D_m. More specifically, for a sufficiently large number of samples m we want to find:

$$h^* = \arg\min_{h \in \mathcal{H}} L_m(h; D_m) \tag{3}$$

Define the generalization error as the difference between the population loss and the empirical loss:

$$\Delta_m(h) = L(h) - L_m(h; D_m) \tag{4}$$

A typical upper bound of the generalization error $\Delta_m(h)$ with probability $1 - \delta$ is given by:

$$\Delta_m(h) \leq \frac{\chi(\mathcal{H}) + O(\log(1/\delta))}{m} \tag{5}$$

If $m >> \chi(\mathcal{H})$ then the generalization error is small with high probability. In the context of deep learning, the number of parameters is typically of the same order as the number of samples or more (the overparametrized regime). Complexity measures such as VC dimension which approximately corresponds to the number of parameters suggest a high generalization error which contradicts empirical evidence that deep learning models can generalize well. This indicates that deep learning models have some implicit bias that enables them to find solution with low effective complexity inducing small generalization error.

- SLT tells us that the generalization error is given by the LLC. The LLC is an effective measure of complexity that is much lower than the VC dimension. SLT applies to degenerate Bayesian models. This indicates that degeneracies are important. Bayesian learning is difficult in practice. Practictioners use SGD. How it does it, how specific function that generalize well are learned is still a mystery. SGD performs well on non-convex high dimensional loss landscapes. Mention Zhang's paper. SGD, the learning algorithm of DeepLearning must do some implicit model selection that enables it to find solutions with low effective complexity (and this implicit model selection presumably differs from Bayesian inference). - Present Deep Linear Networks (DLN) as a setup that we use throughout the survey.

3 Langevin dynamics

It is particularly interesting to figure out the differences between gradient descent and Bayesian inference on high dimensional degenerate loss landscapes.

4 Gradient Descent

Gradient descent is the simplest first order optimization method that was used in DL. To start simple, we look at the implicit biases of GD.

Continuous limit of GD to gradient flow (GF)

Consider a neural network $h(\theta; x, y)$ with parameters $\theta \in \mathbb{R}^d$ and an empirical loss function $L(\theta)$. Let the gradient of the loss be given by $g(\theta) = -\nabla_\theta L(\theta)$. Given some initialization $\theta_0 \sim N(0, \sigma_d^2)$, Gradient Descent is given by the following update rule:

$$\theta_{k+1} = \theta_k - \eta_k g(\theta_k) \tag{6}$$

where η_k is the learning rate with a schedule indexed by k. For theoretical analyses, it is often useful to consider the gradient flow (GF) which is the continuous time limit of gradient descent. We consider that GD is a discretisation of a continuous flow with discrete time steps $t = k\eta$ and let $\eta \to 0$ while being independent of k, then we can write the GF equation:

$$\dot{\theta} := \frac{d\theta}{dt} = -g(\theta, t) \tag{7}$$

- Under what conditions GD continuous limit holds? - Under which conditions GD is stable? - Under what conditions GF converges? - Convergence properties of GD vs GF

Sufficient condition for GD continuous limit to the GF

Consider the Taylor expansion of the parameter θ at time $t + \Delta t$ around time t:

$$\theta(t + \Delta t) = \theta(t) + \dot{\theta}(t)\Delta t + \frac{1}{2}\ddot{\theta}(t)\Delta t^2 + O(\Delta t^3) \tag{8}$$

We use the physicist convention of repeated indices summation. Recall that:

$$\dot{\theta}(t) = -g(\theta, t) \tag{9}$$

$$\ddot{\theta}_i(t) = -\dot{\theta}(t)_j \partial_j \partial_i L(\theta, t) = \partial_{ij} L(\theta, t) \partial_j L(\theta, t) \tag{10}$$

Where the latter equation holds if the loss is $\mathcal{C}^2$. In that case we have:

$$\ddot{\theta}(t) = H(\theta, t) g(\theta, t) \tag{11}$$

Where H is the Hessian of the loss. Assume that $\Delta t = \eta$. Then we have:

$$\theta(t + \eta) = \theta(t) - \eta g(\theta, t) + \frac{1}{2}\eta^2 H(\theta, t) g(\theta, t) + O(\eta^3) \tag{12}$$

Assuming that we have $\eta ||Hg(\theta)|| << 2||g(\theta)||$ then the continuous limit of GD holds. Consider the spectral norm induced by the vector 2-norm i.e.:

$$||H(\theta)||_2 = \sup_{x \neq 0} \frac{||H(\theta)x||}{||x||} = \sigma_{\max}(H(\theta)) \tag{13}$$

Where $\sigma_{\max}(H(\theta))$ is the largest singular value of $H(\theta)$. We have the following inequality:

$$||H(\theta)g(\theta)||_2 \leq ||H(\theta)||_2 ||g(\theta)||_2 \tag{14}$$

Therefore we have:

$$||H(\theta)g(\theta)|| \leq \sigma_{\max}(H(\theta)) ||g(\theta)|| \tag{15}$$

Hence, if $\sigma_{\max} << \frac{2}{\eta}$ then the continuous limit of GD is a valid approximation. Unfortunately, this assumption seems to be violated in practice since it has been observed that GD typically operates at the edge of stability in CIFAR-10 NN[2]. We compared the behavior of the losses for GD and GF in DLNs. **Show experiments here.**

Stability of GD

- Descent lemma: Need to explain as this is the classical picture of stability of GD. Descent lemma gives condition for loss to decrease monotonically. Pb is that with edge of stability this condition is violated but we still observe monotonic loss decrease.

[2]But it might be the case that Gradient flow still converges to good solutions? Or perhaps by studying the second order approximation of GD into the flow things work out? Also not clear if it applies to SGD

Edge of Stability

The Descent lemma gives sufficient condition-the stability condition-for the loss to decrease monotonically. However, it has been observed that deep neural networks, operate at the edge of stability. More specifically, the edge of stability regime follows a phase of progressive sharpening where the sharpness $S(\theta)$ increases monotonically until it reaches the threshold $\frac{2}{\eta}$ and oscillates around that threshold (edge of stability regime).**Insert figure 1 from EoS paper** This shows that GD has an implicit bias induced by the learning rate which regularizes region of the loss landscape whose sharpness is larger than the stability threshold. Mathematically, EoS shows that GD implicitly solving a constrained optimization problem where it minimizes the Loss $L(\theta)$ under the constraint $S(\theta) \leq \frac{2}{\eta}$. We observed this effect in Deep Linear Networks. **Show experiments here.**

We can decompose the GD training dynamics into four phases: a first phase of progressive sharpening, a second phase in which the dynamics blows up along an unstable direction, a third phase of self-stabilization and a fourth phase of returning to stability. Following Damian et al. [2023] we will present a heuristic derivation showing the four phases of GD. We will assume that only one direction u becomes unstable during the dynamics. Let $\theta^\star$ be the point at which the dynamics becomes unstable i.e. the maximum eigenvalue of the Hessian is equal to $\frac{2}{\eta}$. We will assume that only one direction $u := u(\theta^\star)$ becomes unstable during the dynamics. We are interested in the change in sharpness:

$$S(\theta_{t+1}) - S(\theta_t) = \langle \nabla S(\theta_t), (\theta_{t+1} - \theta_t) \rangle = -\eta \langle \nabla S(\theta_t), g(\theta_t) \rangle \tag{16}$$

Let $y_t := \langle \nabla S(\theta^\star), (\theta_{t+1} - \theta_t) \rangle$. Far from the unstable point $\theta^\star$ in the stable regime, we have $S(\theta_t) \leq S(\theta_{t+1})$ i.e gradient of loss and sharpness point in opposite directions. Let us Taylor expand the gradient of the loss around $\theta^\star$:

$$g_i(\theta_t) = g_i(\theta^\star) + \partial_{ij} L(\theta^\star)(\theta_j - \theta_j^\star) + \frac{1}{2}\partial_{ijk} L(\theta^\star)(\theta_j - \theta_j^\star)(\theta_k - \theta_k^\star) + O((\theta_j - \theta_j^\star)^3) \tag{17}$$

One can show that along some stable direction v, when the sharpness is smaller than $\frac{2}{\eta}$ we have:

Hence we can restrict the study of the dynamics along the unstable direction u.

Principal Flow and Central flow

Gradient flow over effective potential

Note that we have the following relation (using symmetry of the Hessian):

$$\nabla^2 L(\theta).\nabla L(\theta) = \frac{1}{2}\nabla(\nabla L(\theta)^T \nabla L(\theta)) \tag{18}$$

Which leads to the following gradient flow (**Comment: Sign mistake?**):

$$\dot{\theta} = -\nabla(L(\theta) - \frac{1}{4}\eta \nabla L(\theta)^T \nabla L(\theta)) = \nabla(L(\theta) - \frac{1}{4}\eta ||\nabla L(\theta)||^2) = -\nabla \tilde{L}(\theta) \tag{19}$$

Where

$$\tilde{L}(\theta) = L(\theta) - \frac{1}{4}\eta ||\nabla L(\theta)||^2$$

In other words, the second order term correction can be incorporated into the gradient flow as an implicit regularization of the loss. This regularization encourage larger gradients i.e. moving toward sharper regions of the loss landscape, which is consistent with observations made on the edge of stability. This regularization was introduced by Barrett and Dherin [2020]. Unfortunately, this second order development is not sufficient to explain the edge of stability phenomena. Note: perhaps we need to go to higher order term to understand the full edge of stability phenomena.

Modes and low rank bias in DLNs

Modes during EoS

Discuss chaotic behavior of modes during EoS in that paper here. EoS can break conservation laws during GF (but that's ok if we can use effective potential).

Implicit Biases of Gradient Descent

The implicit biases of gradient descent have emerged as a fundamental topic in understanding why overparameterized neural networks generalize well despite classical learning theory predictions. This literature review examines the key theoretical advances in characterizing these implicit biases, particularly focusing on recent developments from 2020-2025.

4.0.1 Low-Rank Bias in Deep Linear Networks

The low-rank bias represents one of the most mathematically well-characterized implicit regularization effects in deep learning. For a deep linear network with weight matrices $W_1, W_2, \ldots, W_L$, the learned representation $W = W_1 W_2 \cdots W_L$ exhibits significantly lower effective rank than the ambient dimension would suggest.

Soudry et al. [2018] established foundational theoretical results for linearly separable data with exponential-tailed losses, proving that gradient descent converges to the max-margin solution with precise convergence rates: directional convergence at $O(\log\log(t)/\log(t))$ and loss convergence at $O(1/t)$. This max-margin bias was extended by Lyu and Li [2019] to homogeneous neural networks, demonstrating convergence to KKT points of parameter-space margin maximization.

The nuclear norm minimization conjecture, initially proposed by Gunasekar et al. [2017], suggested that gradient flow with infinitesimal initialization converges to minimum nuclear norm solutions. However, this view was fundamentally challenged by subsequent work. Arora et al. [2019] and Li et al. [2021] showed that matrix factorization exhibits implicit bias toward rank minimization rather than nuclear norm minimization. Specifically, Li et al. [2021] proved that gradient flow on depth-2 matrix factorization is mathematically equivalent to "Greedy Low-Rank Learning" – a sequential rank minimization algorithm that greedily adds rank-1 components.

Recent work by Chou et al. [2024] provided rigorous dynamical analysis characterizing how the effective rank evolves during training in a distinctive "waterfall" pattern, where different eigenvalues converge at different rates. Their analysis reveals that deeper networks exhibit stronger low-rank bias, with the bias strength depending critically on initialization scale and network depth.

4.0.2 Edge of Stability Phenomenon

The edge of stability (EoS) phenomenon, discovered by Cohen et al. [2021], represents a paradigm shift in understanding neural network optimization. During training, the sharpness (largest eigenvalue of the loss Hessian) progressively increases until reaching approximately $2/\eta$, where η is the learning rate, then hovers at this critical threshold while loss continues decreasing.

Damian et al. [2023] provided the first rigorous mathematical explanation through self-stabilization theory. The dynamics can be captured by cubic Taylor expansion: as iterates diverge along the top Hessian eigenvector, the cubic term causes curvature to decrease until stability is restored. This creates a self-stabilizing effect where gradient descent implicitly follows projected gradient descent under the constraint $S(\theta) \leq 2/\eta$.

Arora et al. [2022] demonstrated that EoS represents a novel mechanism of implicit regularization occurring due to non-smooth loss landscapes. The mathematical analysis shows that

gradient descent updates evolve along deterministic flows on the manifold of minimum loss, contrasting with previous results that relied on infinitesimal updates or noise.

Song et al. [2023] extended the theoretical understanding through bifurcation theory, proving that different gradient descent trajectories align on specific bifurcation diagrams independent of initialization. This trajectory alignment provides rigorous mathematical foundations for both progressive sharpening and EoS phenomena. Recent extensions to stochastic settings reveal that SGD operates in "Edge of Stochastic Stability," where batch sharpness rather than full-batch sharpness stabilizes at $2/\eta$.

Milestones: explicit EoS studies in DLNs 2022 - Two-layer linear net [Li et al., 2022]: Four-phase trajectory showing progressive sharpening, EoS oscillation, then eventual monotone descent.

2022 - General GD theory on EoS [Arora et al., 2022]: GD follows a deterministic flow on the low-loss manifold even past the stability bound.

2023 - Diagonal DLNs [Labarrière et al., 2024]: Characterises implicit regularisation at EoS; explains why SGD can benefit from slightly super-critical learning rates.

2023 - Two-step GD beyond EoS [Chen et al., 2023]: Modified updates tame oscillations without shrinking the step-size.

2024 - UV toy model [Kalra et al., 2023]: Minimalist proof that progressive sharpening and EoS appear with a single training example.

2025 - Deep matrix-factorisation [Ghosh et al., 2025]: First fine-grained analysis of loss oscillations in a subspace whose dimension is set by the learning rate.

2025 - Single-neuron-per-layer chains [Yoo et al., 2025]: Relates dataset difficulty and depth to how fast sharpness saturates at the $2/\eta$ threshold.

Historical note: classical work by Saxe et al. [2013] already captured the "progressive sharpening" of singular values in DLNs, but did not frame it in EoS terms.

Insights distilled from the linear literature

- **Sharpness self-regulation.** Exact or bounded spectra show $\lambda_{\max}$ equilibrates near $2/\eta$ once conservation laws linked to layer-wise products break at the threshold [Ghosh et al., 2025].
- **Low-dimensional oscillations.** Beyond the threshold, loss oscillates inside a subspace of dimension $\leq \lceil 2/\eta \rceil$—rank 1 for depth-2 nets [Li et al., 2022].
- **Bifurcation picture.** Fixed η triggers a period-doubling cascade and even chaos in deep linear chains, mirroring nonlinear ReLU behaviour [Ghosh et al., 2025, Kalra et al., 2023].
- **Implicit-bias shift.** At EoS, usual max-margin or minimum-norm biases weaken; diagonal-network analysis shows large η tilts GD and SGD towards different sparsity profiles [Labarrière et al., 2024].
- **Depth–data interplay.** More depth lowers the η needed to hit EoS, while *harder* datasets delay sharpness saturation [Yoo et al., 2025].

Open directions

1. Extending proofs to *non-square* or convolutional linear networks.
2. Linking EoS oscillations to *generalisation*; most current theory still uses synthetic data.
3. Developing a full EoS theory for *mini-batch SGD*; first steps exist but are incomplete [Kalra et al., 2023].

Bottom line. Multiple independent lines of evidence now confirm that DLNs *do* exhibit edge-of-stability dynamics. Because their algebra is explicit, they have become the canonical sandbox for explaining why sharpness hovers near $2/\eta$, how oscillations arise, and what this implies for implicit regularisation.

Additional Implicit Biases

Beyond low-rank bias and edge of stability, neural networks exhibit multiple implicit biases that collectively shape learned representations.

4.0.3 Feature Averaging Bias

Li and Wei [2024] rigorously characterized feature averaging bias, showing that two-layer ReLU networks trained with gradient descent on multi-cluster data converge to averaged combinations of discriminative features rather than learning individual features. This bias creates fundamental adversarial vulnerability, as networks become susceptible to perturbations aligned with negative directions of averaged features.

4.0.4 Spectral Bias

Rahaman et al. [2019] established that neural networks exhibit spectral bias, learning from low to high frequencies during training. The mathematical analysis through Neural Tangent Kernel theory reveals that for harmonic functions $f(x) = \sum a_k \sin(kx)$, low-frequency components (small k) are learned first, with convergence time scaling as $O(\kappa^d/p(x))$ where $p(x)$ is local data density.

4.0.5 Simplicity Bias

Shah et al. [2020] demonstrated that networks exhibit simplicity bias, preferring simple over complex features even when complex features are more predictive. They showed extreme cases where networks become invariant to all predictive complex features, relying exclusively on the simplest available features. Morwani et al. [2023] proved that 1-hidden layer networks depend primarily on linearly separable subspaces, providing mathematical characterization of this bias.

4.0.6 Sparsity Bias

Woodworth et al. [2020] showed that for diagonal networks, initialization scale controls implicit bias: as initialization scale $\alpha \to 0$, the implicit regularizer approaches $R(w) = \|w\|_1$, while $\alpha \to \infty$ gives $R(w) = \|w\|_2^2$, creating automatic feature selection without explicit regularization.

4.0.7 Discrete versus Continuous Optimization

The relationship between gradient descent (discrete updates) and gradient flow (continuous time limit) reveals important distinctions in implicit bias. While gradient flow provides clean theoretical analysis, finite step sizes introduce discretization effects that can significantly alter implicit biases.

For linear models, both formulations generally converge to similar solutions – minimum ℓ_2-norm interpolators for squared loss and max-margin solutions for exponential losses. However, discrete gradient descent with finite learning rates can exhibit different implicit regularization patterns not captured by continuous analysis [Pesme and Flammarion, 2021].

In neural networks, the differences become more pronounced. Homogeneous networks with favorable curvature properties along gradient flow trajectories allow discrete gradient descent to closely approximate continuous flow. However, finite step sizes can induce different implicit

biases, particularly in stochastic variants where different eigenvalue directions are explored [Mulayoff et al., 2021].

4.0.8 Recent Theoretical Advances (2020-2025)

The period 2020-2025 has witnessed several breakthrough theoretical advances. The discovery of Edge of Stability fundamentally changed understanding of neural network optimization, revealing that large learning rates create beneficial implicit bias toward flatter minima through self-stabilization mechanisms.

Feature averaging identification by Li and Wei [2024] provided the first rigorous explanation for adversarial vulnerability in gradient descent, showing that networks trained with standard procedures systematically learn vulnerable representations. This breakthrough suggested that granular supervision (classifying individual features rather than original classes) can achieve optimal adversarial robustness.

Matrix factorization theory experienced major advances with Li et al. [2021] definitively resolving the nuclear norm conjecture, proving that gradient flow implements greedy rank minimization rather than nuclear norm minimization. Chou et al. [2024] provided precise dynamical analysis of how effective rank evolves during training.

Extensions to non-homogeneous networks by Cai et al. [2025] represent significant theoretical generalization, providing asymptotic convergence characterization for generic deep networks beyond the homogeneous case. Ravi et al. [2024] generalized foundational binary classification results to realistic multiclass settings through the framework of Permutation Equivariant and Relative Margin-based (PERM) losses.

4.1 Experiments on Deep Linear Networks

5 Implicit biases of anisotropic noise

5.1 SGD as anisotropic Langevin dynamics (ALD)

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